

REPORT ON INTEGRATED PULL REQUESTS

ABSTRACT. This report summarizes the already integrated pull-request material in the latest high-numbered integration band, roughly PRs #775–#985. It is based on the local integration ledger, committed experimental notes, formalization packages, and representative source notes; it is not a fresh execution replay. The main conclusion is that the PR stream is useful, but mostly as a proof program: it narrows residual cases, repairs false statements, adds formalized kernels, and improves audit discipline. The most promotion-worthy material is the Paper D v13.2 cap package, the Lean discharges for Paper D components, and the KoalaBear M1 rank-nine route cuts. The finite adjacent safe side and the asymptotic entropy-frontier theorem remain open.

1. SCOPE AND METHOD

The request was to look at PRs already integrated, not to merge a new open queue. I therefore used the local repository state, especially `experimental/agents-log.md`, the recent commits ending at `951ad203`, and the integrated notes under `experimental/notes/`. The effective scope is the latest approximately 200-number band of already handled PRs, from the L1/BC/L2 wave around #775 through the M1 moving-root boundary cut #985.

No Python, Lean, Sage, or long-running certificate script was executed for this report. Status words such as `PROVED`, `CONDITIONAL`, `COUNTEREXAMPLE`, `AUDIT`, and `EXPERIMENTAL` are inherited from the integrated notes and should be read subject to their local hypotheses. In particular, “proved local” means proved under the packet’s printed contracts; it does not by itself promote a main-paper theorem or close the Proximity Prize threshold.

2. EXECUTIVE ASSESSMENT

The integrated PRs are not noise. They have turned the repository from a collection of speculative ledgers into a much more structured proof program. The best contributions fall into four groups.

First, the Paper D line has matured. The v13.2 promotion gives a current authority source for the cap paper, with identity-prefix unsafe edges and a clear adjacent-safe certificate template. It supersedes the old v12/raw-v13 split as the main Paper D target.

Second, the formalization stream is now genuinely useful. Several Lean PRs do not merely restate definitions: they refute false skeleton signatures, repair interfaces, and discharge previously sorried kernels. The strongest recent example is the ECFFT constructive discharge: `prop_rational_floor` and `cor_ecfft_onestep` are proved with byte-identical statements in the Lean skeleton. The phi-fiber repair certificate is also valuable because it separates an old Lean-skeleton defect from the actual Paper D statement.

Third, the KoalaBear M1 branch is more concrete. The sequence of rank-nine route cuts now turns a broad full-outside high-degree residual into a small list of named algebraic one-slack cases. The latest cut, PR #985, shows that after the source-rational owner, every surviving full-outside rank-two record has

$$s \geq 67,474, \quad e \geq 33,737, \quad \deg \gcd(P, Q) \leq 1,014,838,$$

and the remaining one-slack shapes are

`UNPAID_NONBASE_COMMON_LINEAR_GCD_TWIST,`
`UNPAID_SPLIT_GCD_NONBASE_LINEAR_MOVING_COFACTOR.`

This is useful because it tells the next proof exactly where to look.

Fourth, the counterexample and audit PRs are doing necessary cleanup. They prevent attractive but false shortcuts: wrong support universes, false strict crossing demands, stale certificate bindings, and overbroad heavy-fiber claims. Those results do not “move the leaderboard”, but they make the proof program less fragile.

3. INTEGRATED WAVES

PR range	Main content	Utility judgment	Promote now?
#775, #777, #779–#802	L1 mixed-petal and CRT packets, BC chart typing, resonant/fold/cylinder reductions, L2 rank-15/rank-16 route cuts, and several Lean threshold compilers.	Useful for the final-resolution atlas. It makes residual cells more checkable, but it does not close Q/BC/SP or any deployed adjacent safe row.	No; summarize selectively.
#803–#815	L1 shared-auxiliary and total-degree CRT packets, rank-cell exclusions, KoalaBear base-slope universe audits, BC deficiency-2 typing, and site updates for corrected rows.	High utility for finite ledgers and public-board accuracy. The most visible effect is better row display, not a new theorem closing the main gap.	Site yes; paper selective.
#817–#845	Threshold/compiler notes, C0 periodic Lean stack, M1 branch stack, L2 source certificates, fixed-slope Johnson formalization, and rank-one corrections.	Important infrastructure. The Lean and compiler items are reusable, while many route cuts remain local and conditional on packet contracts.	Map only.
#849, #867, #871, #874	KoalaBear branch-3 rank-nine route-cut stack: low-excess carrier cut, actual core MDS rank ladder, mask-deficit route cut, and syndrome-rank reduction.	High utility for M1. These packets make the rank-nine branch structured, but leave sparse-sigma/rank-nine residuals and the KoalaBear row open.	No; keep as route cuts.
#879–#880	Paving-v8 audit and dense-shell class-charge material.	Useful for ePrint readiness and sign/class-sum separation. It is supportive for the paving paper rather than a direct final-threshold proof.	Audit yes; theorem no.
#881–#950	Large wave: Route-D no-go and recursion tools, dense-shell repairs, transfer-shape evidence, M1 rank-nine route cuts, L2 rank-16 fixed-26/fixed-27 cuts, Paving retained-factor audits, and Lean progress.	Mixed but useful. The important effect is a stronger residual map and better guardrails against false promotion. Heavy computations should remain off-by-default.	No wholesale promotion.
#951–#975	Route-D owner-typed forest and marked recursion, M1 source-Mobius and projective-base refinements, L2 fixed-26/fixed-27 local packets, dense-shell certificate audits, affine-line Steiner quotient ownership, and Lean repairs.	Very useful locally. PR #962 is the strongest mathematical move in this subwave, removing a selector-provenance gate for one M1 full-outside maximal-gcd cell. PR #965 is important because it keeps stale Arb evidence in OPEN-GAP status.	Only audited summaries.
#976–#979	Affine-prefix fixed-point and pole strata, primitive top-seam marked-incidence audit, and complete-fiber subset-packing Lean package.	Good regression material. The top-seam audit is especially useful against marked-incidence overcounting; the Lean subset-packing compiler is reusable. No leaderboard movement.	Lean maybe; theory no.
#980–#984	Phi-fiber Lean repair certificate, ECFFT Lean constructive discharge, O5c strict-crossing counterexample, Case-B support-universe correction, and M1 source-rational owner route cut.	High value. The Lean discharges and phi-fiber repair are audit/formalization wins; the O5c and Case-B notes prevent wrong proof targets; the M1 owner cut narrows the KoalaBear rank-nine residual.	Audit.

PR range	Main content	Utility judgment	Promote now?
#985	M1 moving-root cofactor/slack and C5 boundary cut.	High local utility. It raises the post-#982 full-outside source floor to $s \geq 67,474$ and leaves two explicit one-slack residuals.	Route-cut summary yes.

4. MOST USEFUL RESULTS

4.1. Paper D and threshold authority. The main paper-level movement is not an individual small PR but the promotion of `tex/cs25_cap_v13_2.tex`. It makes Paper D v13.2 the current authority source. Its useful contribution is the status split:

unsafe exact certificates are theorem-level in the paper package, while adjacent first-safe statements remain conditional on exact upper ledgers. For the deployed KoalaBear MCA row, the current experimental unsafe agreement is $a_0 = 1,116,047$, corresponding to

$$\delta = \frac{981105}{2097152} \approx 0.4678273.$$

The next finite target is not to get a bigger error at a worse radius; it is to prove the adjacent safe certificate at $a_0 + 1 = 1,116,048$, with exact integer constants.

4.2. Formalization. The integrated Lean material has crossed from outline to useful proof infrastructure. The most important formalization items are:

- PR #961, which refutes false Paper-D-v12 skeleton statements, repairs the formulations, and proves the $c = 1$ first-grid cap chain.
- PR #977, which formalizes the complete-fiber subset-packing compiler.
- PR #980, which gives a Lean counterexample to the old phi-fiber skeleton signature and a coefficient-to-value wrapper matching the Paper D v13.2 premise $\varphi \in B[X]$.
- PR #983, which proves `prop_rational_floor` and `cor_ecfft_onestep` in the ECFFT skeleton, reducing the documented sorry count.

These are among the most promotion-worthy PR outputs because they reduce ambiguity rather than adding another heuristic layer.

4.3. M1 KoalaBear safe-side route cuts. The M1 packets are the strongest route toward the finite adjacent safe side. Their value is cumulative. Early branch-3 packets split the problem into rank and carrier cases. The source-Mobius/source-rational packets then remove large projective-rational owner classes without increasing the paid bound. The moving-root C5 boundary packet composes the cofactor divisibility with the source-rational survivor inequality.

The important current local endpoint is:

$$\text{post-C5 full-outside rank-two survivors have } s \geq 67,474, e \geq 33,737.$$

This replaces a vague high-degree residual by two explicit one-slack components. The next useful theorem is therefore sharply stated: classify those two nonbase linear components against the regular locator equations and the eight rank-nine outlier directions. Generic high-degree elimination is not the next best move.

4.4. L1, BC, L2, and Route-D. The L1/BC/L2/Route-D material is broad. Its most useful function is not a single theorem but a better residual atlas. The PRs add mixed-petal and CRT evidence on the L1 side, BC chart typing and first-interior modular-fiber notes, fixed-slope and complete-fiber L2 compilers, and Route-D owner/no-go packets.

The judgment is conservative: these should not be promoted wholesale into the main TeX papers. They are best used as a menu of local lemmas and counterexamples. A future paper update should extract only statements that either close a named residual branch or give a reusable theorem with clearly printed hypotheses.

4.5. Audits and counterexamples. Several integrated PRs are valuable because they say “do not prove this”:

- PR #981 refutes a universal O5c strict-crossing demand using a sparse exact-deep family where the exact numerator is 2, equal to or below the frozen target in the relevant endpoint.
- PR #984 corrects a Case-B support-universe mistake: supports must be counted by $\binom{|D|}{m}$, not by $\binom{|B|}{m}$. The corrected fixed-density regime still gives an exponential-field lower picture, but the old $m = 2e$ calibration was wrong.
- PR #978 is a useful primitive top-seam regression against marked incidence overcounting and common-core erasure.
- PR #965 is useful precisely because it refuses to promote stale Arb evidence to a proved transfer theorem.

These negative or audit results should stay visible. They prevent future agents from spending cycles on false shortcuts.

5. WHAT SHOULD BE PROMOTED

The following items are strong candidates for promotion into a clean TeX writeup, after human review.

- (1) The ECFFT Lean discharge can be summarized in the Paper D formalization appendix or an audit note: the rational floor and one-step ECFFT cap have machine-checked skeleton proofs with byte-identical statements.
- (2) The phi-fiber repair certificate should be cited in the Paper D audit map as a formal check that the base-field polynomial premise is load-bearing and correctly bridges to the repaired value-level Lean theorem.
- (3) The M1 KoalaBear route-cut sequence, ending with PR #985, should be summarized as the current best finite safe-side residual reduction. It should not be advertised as KoalaBear closure.
- (4) The Case-B correction should be summarized in the asymptotic/frontier notes because it changes the correct support-counting calibration and separates subexponential-target lower constructions from fixed-epsilon constant-fraction questions.
- (5) The complete-fiber subset-packing Lean compiler from PR #977 should be kept in the formalization roadmap as reusable L2 infrastructure.

6. WHAT SHOULD NOT BE PROMOTED YET

The following should remain experimental.

- (1) Local route cuts whose hypotheses depend on selector provenance, first-match owner order, or frozen packet contracts.
- (2) Large Python/Sage certificate packets unless a maintainer explicitly requests replay and the certificate is source-bound.
- (3) Leaderboard or site rows that are only evidence, not theorem-level unsafe/safe certificates.
- (4) Broad L1/BC/L2/Route-D stacks that narrow residuals but do not close a named theorem.
- (5) Any statement using unspecified polynomial losses to close a finite adjacent deployed row. The finite rows need constants fitting inside the printed bit margin.

7. CURRENT RESEARCH STATE AFTER THESE PRS

For the finite deployed Proximity Prize direction, the unsafe side is much sharper than before. The main remaining task is the adjacent safe certificate: if $B^* = \lfloor 2^{-128} q_{\text{line}} \rfloor$, prove

$$U(a_0 + 1) \leq B^* < L(a_0)$$

with an exhaustive, deduplicated upper ledger U and an exact lower staircase L . The PRs mostly improve U , especially through M1 route cuts.

For the asymptotic direction, the conjectural picture is still the entropy-subfield frontier

$$\delta_C^*(\varepsilon^*) = 1 - \rho - g^*(\rho, \log_2 |B|) + o(1),$$

where g^* is determined by the binary-entropy balance. The recent PRs do not prove this theorem, but they improve the evidence and remove several false formulations of the required support-counting and residual-payment inputs.

For formalization, the project is clearly moving in the right direction. The next best formalization work is not to add more skeletons, but to mechanize the exact staircase compilers, the Paper D v13.2 cap chain, and the M1 route-cut statements with their dependencies explicitly printed.

8. BOTTOM LINE

The last integrated PR band is useful and worth keeping, but most of it should remain in **experimental/**. The material most likely to survive into a submission-facing paper is:

Paper D v13.2 authority + Lean formalization discharges

+ KoalaBear M1 residual route cuts + support-universe and strict-crossing audits.

The project is closer to RS MCA because the remaining proof obligations are now narrower and better named. It is not done: the finite adjacent safe certificate and the asymptotic entropy-frontier upper theorem still require new proof, not just more certificates.